

Exact sign pairing in the permanent formula

Proof of Conjecture 00000006322

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Abstract

For an $n \times n$ matrix with $n \geq 1$, we derive the full 2^n -term signed product-of-sums formula for the permanent and prove that its terms pair under simultaneous sign reversal. Each pair contributes twice either representative. Fixing one coordinate to $+1$ leaves exactly 2^{n-1} terms, with normalization $2^{-(n-1)}$. The Lean proof covers every positive matrix size and every field in which $2 \neq 0$, and uses Mathlib's standard permanent.

Precise statement and conventions

Let I be a finite set of size $n \geq 1$, let R be a field with $2 \neq 0$, and let $A = (a_{ij})_{i,j \in I}$ be a square matrix over R . In particular the result applies over the rationals, reals or complex numbers. Define

$$\text{per}(A) = \sum_{\sigma \in \text{Sym}(I)} \prod_{j \in I} a_{\sigma(j),j}.$$

For a sign vector $\delta \in \{1, -1\}^I$ put

$$W_A(\delta) = \left(\prod_{i \in I} \delta_i \right) \prod_{j \in I} \left(\sum_{i \in I} \delta_i a_{ij} \right).$$

Choose any coordinate $i_0 \in I$. We prove the following exact form of the source's pairing and conversion assertions.

Theorem 1. *For every such matrix,*

$$\text{per}(A) = \frac{1}{2^{n-1}} \sum_{\substack{\delta \in \{1, -1\}^I \\ \delta_{i_0} = 1}} W_A(\delta).$$

The displayed sum has exactly 2^{n-1} indexed terms. It is obtained from the full 2^n -term formula by the fixed-point-free pairing $\delta \leftrightarrow -\delta$, and paired terms are equal.

This is the classical sign-sum permanent identity commonly called Glynn's formula; a published elementary derivation is given by Spies. We supply the full proof and a general Lean formalization. The term count describes this particular representation, not a lower bound on all algorithms or a claim that every summand is nonzero for every matrix. Positive size is required for the free pairing. The empty permanent has its separate convention 1.

Derivation of the full formula

Distributing the n column sums gives

$$W_A(\delta) = \sum_{f: I \rightarrow I} C_f(\delta) \prod_{j \in I} a_{f(j),j}, \quad C_f(\delta) = \left(\prod_{i \in I} \delta_i \right) \left(\prod_{j \in I} \delta_{f(j)} \right).$$

We evaluate the sum of C_f over all sign vectors.

If f is bijective, then $\prod_j \delta_{f(j)} = \prod_i \delta_i$. Therefore $C_f(\delta) = 1$ for every sign vector, since $\delta_i^2 = 1$. Its sum is 2^n .

If f is not bijective, finiteness of its common domain and codomain implies that f is not surjective. Choose $r \notin f(I)$. Flipping only δ_r negates the first product in C_f and leaves the second unchanged. This flip is an involutive bijection of the sign vectors. Thus its sum S_f satisfies $S_f = -S_f$; since $2 \neq 0$ in a field, $S_f = 0$.

Interchanging the two finite sums consequently yields

$$\begin{aligned} \sum_{\delta \in \{1, -1\}^I} W_A(\delta) &= \sum_{f: I \rightarrow I} \left(\sum_{\delta} C_f(\delta) \right) \prod_{j \in I} a_{f(j), j} \\ &= 2^n \sum_{\sigma \in \text{Sym}(I)} \prod_{j \in I} a_{\sigma(j), j} = 2^n \text{per}(A). \end{aligned}$$

This proves the full formula directly from the permutation definition; the full formula is not an additional assumption.

The pairing and its exact constant

Simultaneously negating all coordinates contributes $(-1)^n$ from the prefactor and another $(-1)^n$ from the n column sums. Consequently

$$W_A(-\delta) = (-1)^{2n} W_A(\delta) = W_A(\delta).$$

Because $n \geq 1$ and $1 \neq -1$, no sign vector equals its negative. Every two-element pair has exactly one member with $\delta_{i_0} = 1$. These representatives are indexed freely by the other $n - 1$ signs, so there are exactly 2^{n-1} of them. It follows that

$$2^n \text{per}(A) = \sum_{\delta} W_A(\delta) = 2 \sum_{\delta_{i_0}=1} W_A(\delta).$$

Cancel 2 and then divide by the nonzero 2^{n-1} to obtain Theorem 1. Thus the normalization changes from 2^{-n} to $2^{-(n-1)}$, exactly compensating for retaining one representative of each equal pair. This proves all three pairing, constant and term-count assertions in the stated formula.

Formal correspondence

The Lean result is generic over a finite index type, a field R , and the assumption `NeZero` ($2 : R$). The entries of the matrix are arbitrary. `Matrix.permanent` is used directly. Signs are encoded by Booleans: false is +1 and true is -1. The theorem `sign_injective` verifies that these are distinct field values under the stated assumption.

The theorem `term_expansion` distributes the column sums; `coefficient_sum` proves the cancellation for every nonbijective row-choice function. An explicit equivalence between bijective functions and permutations connects the surviving terms to the actual permanent in `full_formula`.

For the halving step, `opposite_involutive` and `opposite_ne_self` establish the free pairing. `term_opposite` proves equality of its summands, and `half_sums_equal` constructs the bijection between the two halves. `number_of_terms` proves the cardinality $2^{|I|-1}$. The final theorem `pairing_reduction` combines the free pairing, the factor two, the exact count and the permanent identity. `glynn_succ` specializes it to every matrix of size $n + 1$, including the one-by-one case.

Reproduction

Lean 4.19.0 and Mathlib revision

c44e0c8ee63ca166450922a373c7409c5d26b00b are pinned. In `lean/`, run

```
lake exe cache get
```

```
lake build
```

```
lake env lean -DwarningAsError=true Main.lean
```

The submitted project is rebuilt in a fresh directory using only verified official dependency artifacts. No numerical test family substitutes for the general proof. There are no proof placeholders, custom axioms or external computational oracles. No auxiliary numerical program is required. Run `tectonic main.tex` to regenerate the PDF. The author performed a separate adversarial self-review; no independent reviewer or subagent was used.

References

1. The Last Math Competition, Conjecture 00000006322. The exact bilingual source and its upstream provenance are included.
2. J. Spies, *A formula for the permanent*, Nieuw Archief voor Wiskunde 5/21(1) (2020), p. 27. Publisher's article.